

12(a)(i)	fission	B1
12(a)(ii)	<i>either</i> ${}^0_{-1}\text{e}$ <i>or</i> ${}^0_{-1}\beta$	M1
	7	A1
12(b)(i)	energy = $c^2 \Delta m$ = $0.223 \times 1.66 \times 10^{-27} \times (3.00 \times 10^8)^2$	C1
	= $3.33 \times 10^{-11} \text{ J}$	A1

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Question	Answer	Marks
12(b)(ii)	Any 2 from: kinetic energy of products gamma photons neutrinos	B2